

Foundations of LLMs & Cloud Computing Basics

Overview

14 Aug 2026

- **establish foundation in LLM architecture**
 1. **describe the key differences between major LLM providers**
 - i. **OpenAI**
 - ii. **Anthropic**
 - iii. **Google**
- **set up and configure cloud development environments for AI development**
 1. **make your first API calls through the UNC AI Gateway**
 - i. **Azure**
 - ii. **OpenAI**
- **reinforce understanding of transformers**
 1. **explain the self-attention mechanism and its role in transformer architecture**

Required Readings

- Vaswani, A., et al. (2017). "Attention Is All You Need."
 - the foundational transformer paper
 - focus on Sections 1-3 and the attention mechanism diagrams
 - <https://arxiv.org/abs/1706.03762>
- Brown, T. B., et al. (2020). "Language Models are Few-Shot Learners."
 - the GPT-3 paper that introduced modern prompting
 - read the introduction and emergent abilities section
 - <https://arxiv.org/abs/2005.14165>
- OpenAI (2023). "GPT-4 Technical Report."
 - understand current capabilities and limitations

- <https://arxiv.org/abs/2303.08774>

Required Videos

- Understanding Transformer Architecture (4 min)
- Understanding Attention Mechanisms (4 min)
- Introduction to Cloud Platforms (3 min)
- Overview of Current Model Landscape (5 min)

Supplementary Videos

- Andrej Karpathy - "Let's build GPT: from scratch" (2 hours)
 - Highly recommended for deep understanding
 - <https://www.youtube.com/watch?v=kCc8FmEb1nY>
- MIT 6.S191 - "Transformers and Attention" (58 minutes) ^{*404}

Cloud Environment Setup

Environment Setup

- get your UNC AI Gateway API key
 - sent via Canvas Inbox
- confirm access to the course models
 - gpt-4.1-mini
 - gpt-5.6-sol
 - text-embedding-3-small
- configure API keys and environment variables

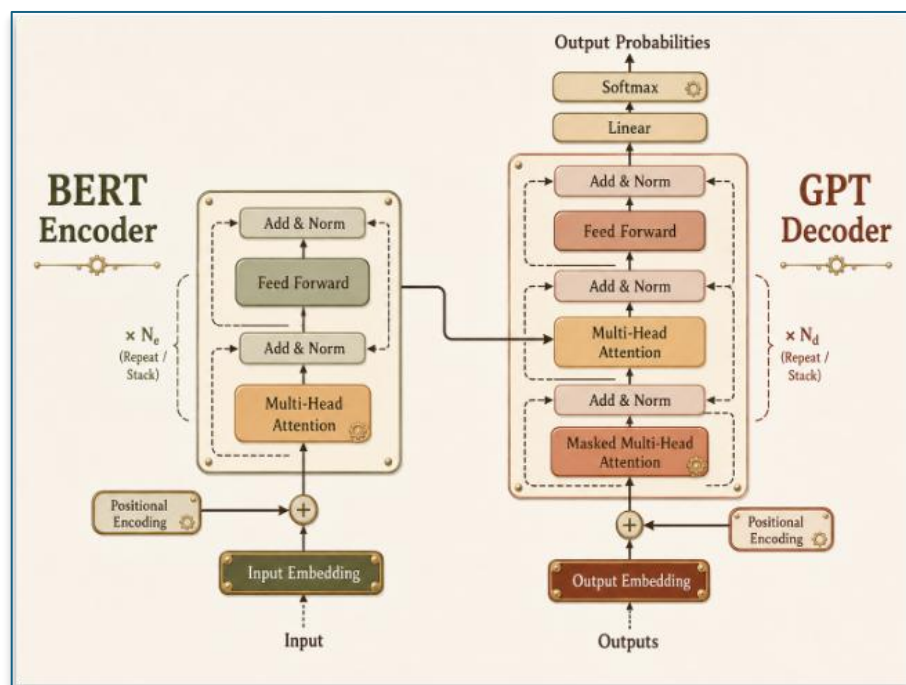
First API Calls

- complete the provided Jupyter notebook
- implement token counting and cost estimation
- deploy basic LLM with security checks

Documentation	• document your setup process • note any challenges encountered
Live Session Preparation	
Planned Discussion Items	• transformer architecture • challenges from the lab • bring questions about the readings
Key Concepts	• self-attention ○ how models attend to different parts of the input simultaneously • tokenization ○ how text is converted to numerical representations • context window ○ maximum input size models can process • temperature ○ controls randomness in generation

Async**Understanding Transformer Architecture****Encoder-
Decoder
Structure**

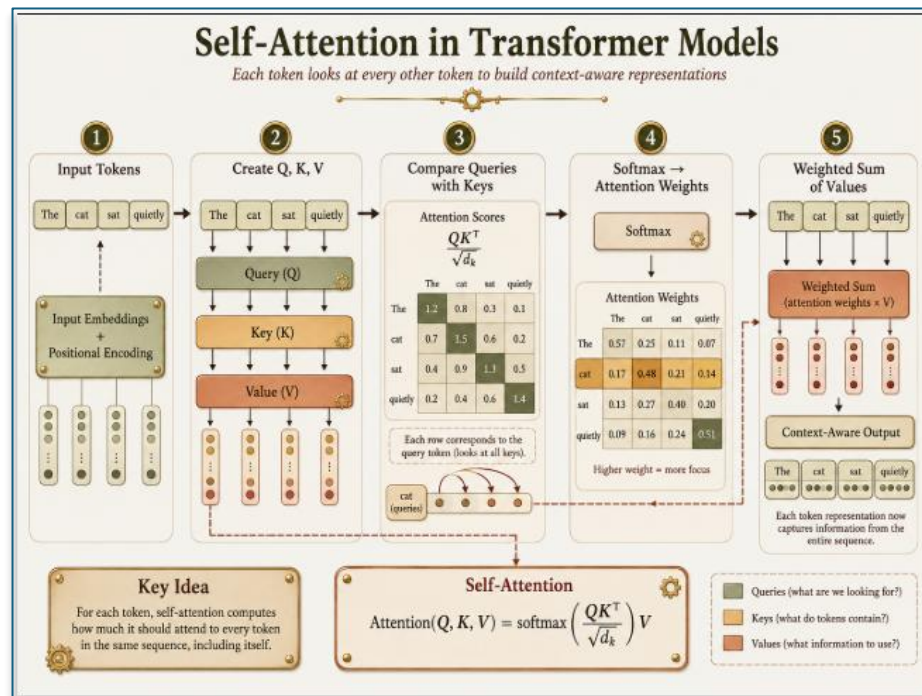
- **Encoder**
 - a stack of identical layers
 - reads input sequence and generates a rich contextual representation
 - a set of context vectors
- **Decoder**
 - also a stack of layers
 - takes the encoders output
 - generates output sequence one token at a time
 - looks at what is already generated to inform the next word



ChatGPT 5.6 Sol

Self-Attention Mechanism

- allows the model to look at all other words in the input sequence
- provides added context for the current word
- weighs the importance of all other words relative to a specific word
- allows the model to capture complex, long-range dependencies and context



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Understanding Attention Mechanisms

Query, Key, Value Vectors

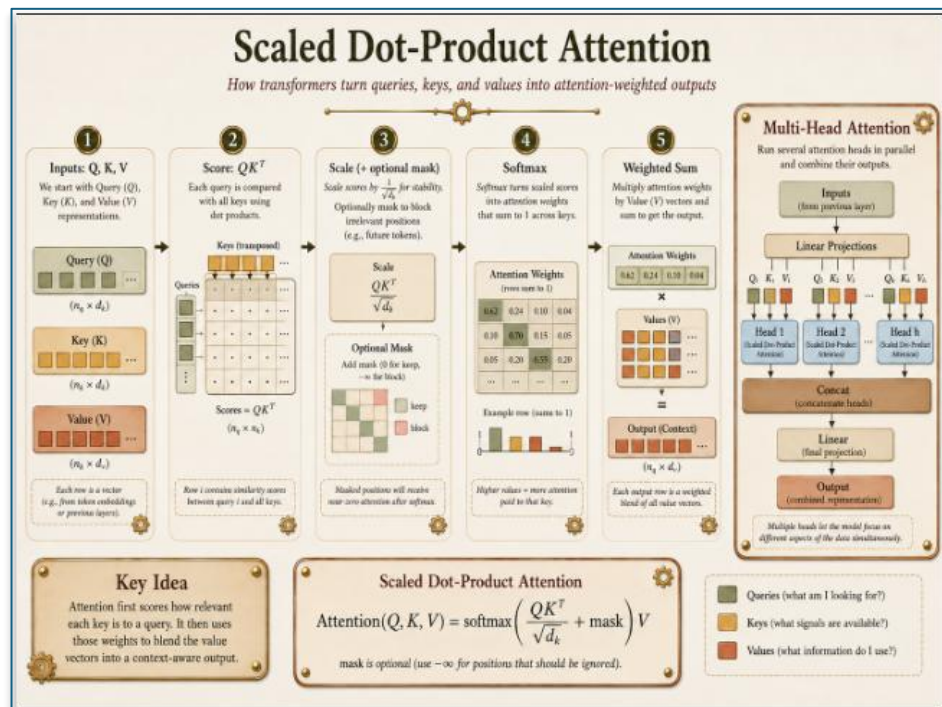
- Query (Q)
 - the current word's question
 - "what am I looking for?"
- Key (K)
 - the "label" of all words in the sequence
 - "what information do I hold?"
- Value (V)
 - the actual content / meaning of the words

Score Calculation

- the mechanism calculates the score by matching the Q of each word with the K of another
- similar to search engine matching your query (Q) string to a video title (K) to return a video (V)

Scaled Dot-Product Attention

- score
 - compute the dot product of the Query with all Keys
- scale
 - divide scores by the square root of the key's dimension (d_k) to stabilize gradients
- softmax
 - normalizes all weights to sum to 1
- output
 - multiply the weights by the Value vectors and sum



ChatGPT 5.6 Sol

Introduction to Cloud Platforms	
IaaS	Infrastructure • delivers on-demand infrastructure ○ compute ○ storage ○ networking • user manages ○ OS ○ middleware ○ applications
PaaS	Platform • provides development platform • the cloud provider manages ○ underlying hardware ○ software resources
SaaS	Software • delivers a complete, cloud-based application • provider manages everything
Major Cloud Providers	• Amazon Web Services (AWS) ○ the market leader ○ extensive and mature catalog of services • Microsoft Azure ○ strong integration with Microsoft's Enterprise ecosystem ○ powerful hybrid cloud story

	- Google Cloud Platform (GCP) - leader in AI, machine learning, and data analytics - leverages Google’s extensive infrastructure		
Model Landscape (video data is stale)			
Model Capabilities	GPT-4.0	Claude Sonnet 3.5	Gemini 1.5 Pro
	- fast all-rounder - native multimodal - text - audio - vision - ideal for real-time conversational AI	- reasoning specialist - leads in - top-tier reasoning - instruction following - coding benchmarks - excellent for complex tasks	“Context King” - good multimodal skills - its defining feature is its massive context window for large-scale analysis
Context Windows	128K	200K	1M